

Shell Scripting

Ahmed Ehab Ahmed

Lab-3

1. Write a script called mycase, using the case utility to checks the type of character entered by a user:

a) Upper Case. b)Lower Case. c)Number. d)Nothing.

```
Lab3 > mycase.sh
1  #!/usr/bin/bash
2
3  read -p "Enter Character: " char
4
5  case $char in
6  [A-Z])
7      echo "Upper Case"
8      ;;
9  [a-z])
10     echo "Lower Case"
11     ;;
12  [0-9])
13     echo "Number"
14     ;;
15  *)
16     echo "Nothing"
17     ;;
18  *)
19     echo "Other Character"
20     ;;
21  esac
22
```

2. Enhanced the previous script, by checking the type of string entered by a user:

a. Upper Cases. b)Lower Cases. c)Numbers. d)Mix. e)Nothing.

```
Lab3 > mycase.sh
1  #!/usr/bin/bash
2
3  LC_COLLATE=C
4  shopt -s extglob
5
6  read -p "Enter String: " str
7
8  case $str in
9  +([a-z]))
10     echo "Small characters"
11     ;;
12  +([A-Z]))
13     echo "Capital characters"
14     ;;
15  +([0-9]))
16     echo "Number"
17     ;;
18  +([A-Za-z]))
19     echo "Mixed characters"
20     ;;
21  +([A-Za-z0-9]))
22     echo "Mixed characters and Number"
23     ;;
24  *)
25     echo "Nothing"
26     ;;
27  *)
28     echo "Special Characters"
29     ;;
30  esac
31
```

3. Write a script called mychmod using for utility to give execute permission to all files and directories in your home directory.

```
Lab3 > mychmod.sh
1  #!/usr/bin/bash
2
3  for item in $HOME/*; do
4      chmod +x "$item"
5  done
```

4. Write a script called mybackup using for utility to create a backup of only files in your home directory.

```
Lab3 > mybackup.sh
1  #!/usr/bin/bash
2
3  mkdir -p $HOME/backup
4  for file in $HOME/*; do
5      if [ -f "$file" ]; then
6          cp "$file" $HOME/backup/
7      fi
8  done
```

5. Write a script called mymail using for utility to send a mail to all users in the system. Note: write the mail body in a file called mtemplate.

```
Lab3 > mymail.sh
1  #!/usr/bin/bash
2
3  users=$(cut -d: -f1 /etc/passwd)
4  for user in $users; do
5      mail -s "Notification" "$user" <mtemplate
6  done
```

6. Write a script called chkmail to check for new mails every 10 seconds. Note: mails are saved in /var/mail/username.

```
Lab3 > chkmail.sh
1  #!/usr/bin/bash
2
3  mailfile="/var/mail/$USER"
4  while true; do
5      if [ -s "$mailfile" ]; then
6          echo "You have new mail!"
7      else
8          echo "No new mail."
9      fi
10     sleep 10
11 done
```

Bonus: Open a talk session to a certain user when she/he logs into the system.

```
Lab3 > talk.sh
1  #!/usr/bin/bash
2
3  echo "Enter username to talk to:"
4  read target_user
5  echo "Waiting for $target_user to log in..."
6  while ! who | grep -q "^$target_user"; do
7      sleep 5
8  done
9  talk "$target_user"
```

7. What is the output of the following script

```
typeset -i n1
typeset -i n2
n1=1
n2=1
while test $n1 -eq $n2; do
    n2=$((n2+1))
    print $n1
    if [ $n1 -gt $n2 ]; then
        break
    else
        continue
    fi
    n1=$((n1+1))
    print $n2
done
```

Output: 1

8. Create the following menu:

- a. Press 1 to ls
- b. Press 2 to ls -a
- c. Press 3 to exit

Using select utility then while utility.

```
Lab3 > mymenu2.sh
1  #!/usr/bin/bash
2
3  menu="
4  1) ls
5  2) ls -a
6  3) exit
7  "
8  while true; do
9      echo "$menu"
10     read -p "Choose an option: " choice
11
12     case $choice in
13         1)
14             ls
15             ;;
16         2)
17             ls -a
18             ;;
19         3)
20             break
21             ;;
22         *)
23             echo "Invalid option: $choice. Please try again."
24             ;;
25     esac
26 done
27
```

```
Lab3 > mymenu.sh
1  #!/usr/bin/bash
2
3  PS3="Choose an option: "
4  options=("ls" "ls -a" "exit")
5  select opt in "${options[@]}; do
6      case $opt in
7          "ls") ls ;;
8          "ls -a") ls -a ;;
9          "exit") break ;;
10         *) echo "Invalid option" ;;
11     esac
12 done
13
```

9. Write a script called myarr that ask a user how many elements he wants to enter in an array, fill the array and then print it.

```
Lab3 > myarr.sh
1  #!/usr/bin/bash
2
3  read -p "How many elements? " n
4  for ((i = 0; i < n; i++)); do
5      read -p "Enter element $i: " arr[$i]
6  done
7  echo "Array elements: ${arr[*]}"
8
```

10. Write a script called myavg that calculate average of all numbers entered by a user.
Note: use arrays

```
Lab3 > myavg.sh
1  #!/usr/bin/bash
2
3  read -p "Enter numbers separated by space: " -a arr
4  sum=0
5  count=${#arr[@]}
6  for i in "${arr[@]"; do
7      sum=$((sum + i))
8  done
9  avg=$((echo "scale=2; $sum / $count" | bc))
10 echo "Average is: $avg"
11
```

11. Write a function called mysq that calculate square if its argument.

```
Lab3 > mysq.sh
1  #!/usr/bin/bash
2
3  mysq() {
4      if [ -z "$1" ]; then
5          echo "Please provide a number."
6      else
7          echo $((($1 * $1))
8      fi
9  }
10 mysq $1
11
```